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REFV-KG Explorer

Causal and Stakeholder-Oriented Exploration of Renewable Energy Policy Discussions

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Demo & materials

Explore causes, effects & countermeasures, compare who said what, and check the source passage within one knowledge graph.

523 Japanese policy documents

22,922 causal triples

79.6% of causal triples link to ≥1 stakeholder opinion

1 Problem & demo scenario

Residents, governments, developers and researchers discuss renewable energy projects across council minutes, committee records, government publications, news articles and stakeholder documents.

Scenario: A resident concerned about a planned wind farm searches for wind power and health or noise. The Explorer presents causal statements, stakeholder opinions and supporting passages, with a graph overview to support comparison.

Design requirements

R1 · Explore causal structures
Cause, effect and countermeasure relations around a policy issue, with an overview of how statements interconnect.

R2 · Compare opinions & evidence
For each statement: who expressed related opinions, what they said, and the supporting passage.

2 System overview: data model and exploration workflow

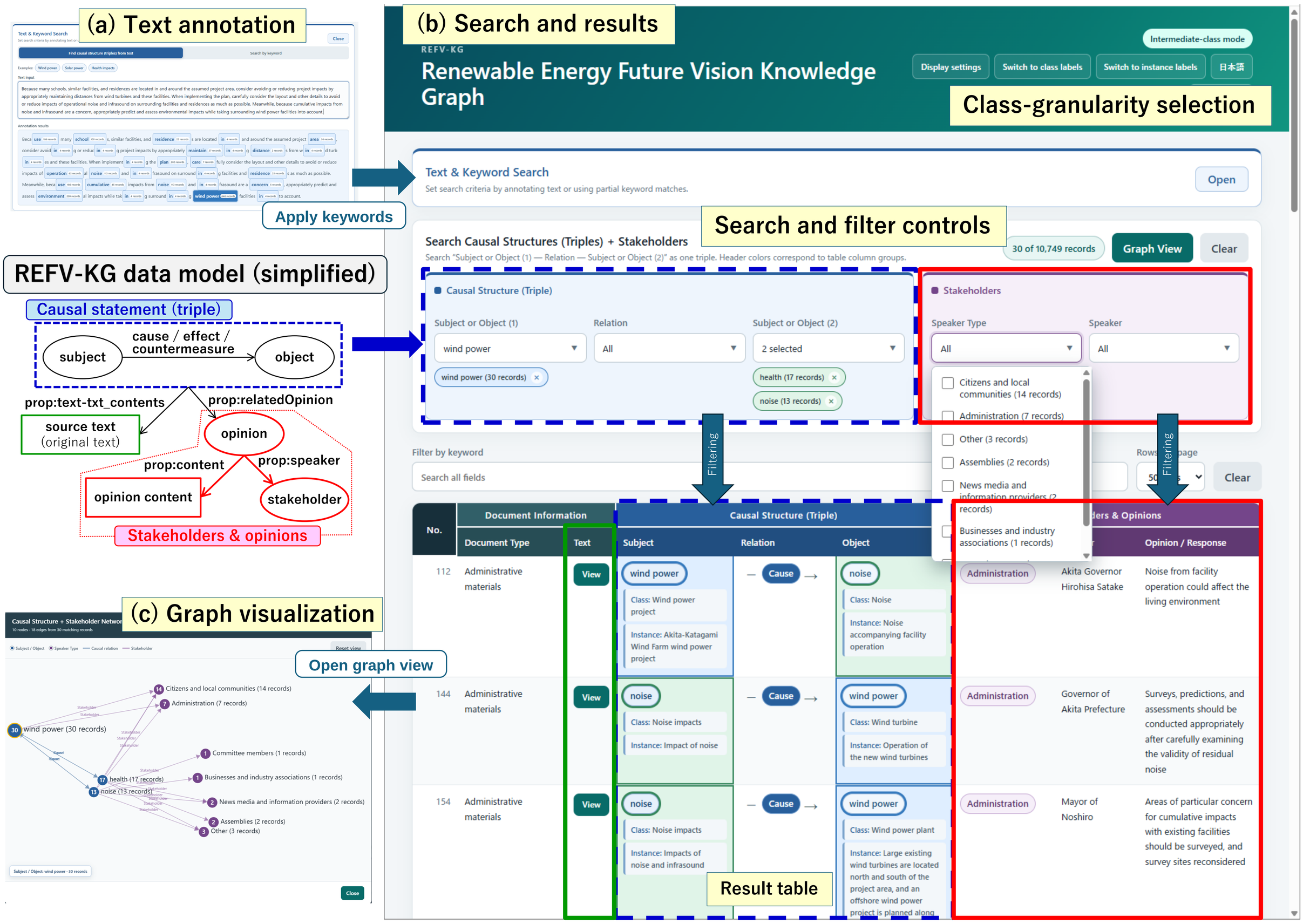


Fig. 1 Matching colors link the data model to the interface: causal statements (blue), source text (green), and stakeholders and opinions (red). Text annotation supplies search keywords. Graph View summarizes the filtered records.

a Text annotation

Annotate discussion text with KG keywords, then apply selected keywords to the search.

b Search & results

Filter by keywords, relation and stakeholder. Each row links a triple to opinions and evidence.

c Graph View

View classes and record counts. Select a node to filter the result table.

Keyword granularity

Search candidates for "wind":

7	53	200+
intermediate (default)	original classes	keyword labels

Class organization

Japanese prefixes group 7,832 keyword classes into 1,369 intermediate classes.

AI-assisted mapping organizes 169 stakeholder classes into 10 types in 5 categories.

Expert feedback informed this design. Original classes remain available.

Limitations & next steps

Intermediate classes need systematic validation.

Transfer to other KGs requires adapting the SPARQL-to-JSON layer and remains unvalidated.

A task-based user study remains future work.

Links

Code: github.com/oecu-kozaki-lab/REFV-KG-Explorer

Demo: refv-ex.kozaki-lab.jp/?lang=en

Data: github.com/oecu-kozaki-lab/REFV-KG

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